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Excitations of a two-dimensional supersolid

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We present a theoretical study of the excitations of the two-dimensional supersolid state of a Bose-Einstein condensate with either dipole-dipole interactions or soft-core interactions. This supersolid state has three gapless excitation branches arising from the spontaneously broken continuous symmetries. Two of these branches are related to longitudinal sound waves, similar to those in one-dimensional supersolids. The third branch is a transverse wave arising from the non-zero shear modulus of the two-dimensional crystal. We present the results of numerical calculations for the excitations and dynamic structure factor characterising the density fluctuations, and study their behavior across the discontinuous superfluid to supersolid transition. We show that the speeds of sound are described by a hydrodynamic theory that incorporates generalized elastic parameters, including the shear modulus. Furthermore, we establish that dipolar and soft-core supersolids manifest distinct characteristics, falling into the bulk incompressible and rigid lattice limits, respectively.

I. INTRODUCTION

A supersolid is a state of matter in which phase and translational symmetries are broken. Here we refer to a D-dimensional supersolid as being a system that spontaneously develops a D-dimensional crystal structure whilst still exhibiting superfluidity. For this system it is expected that the broken symmetries will lead to $(D+1)$-Nambu-Goldstone modes that manifest as gapless excitation branches [1]. The experimental production of supersolid states in atomic gases [2–6] has generated interest in their properties, including their excitation spectra. The $D = 1$ case [7–13] has two gapless excitation branches of longitudinal character, which are referred to as first and second sound[1]. A $D = 2$ supersolid has recently been been produced using a dipolar Bose gas in an oblate shaped trapping potential [16, 17]. This system has an rich phase diagram with different ground state crystal patterns separated by first order transitions [18–22], and is an interesting system for considering the interplay of crystalline order with vortices [23–26].

Here we study the excitation spectrum of a $D = 2$ supersolid. We consider the case of a zero-temperature gas in the thermodynamic limit, which admits a well-defined band structure allowing us to investigate the gapless modes in detail. We show that a hydrodynamic theory provides a precise description of the excitations in terms of a set of underlying elastic parameters. Supersolids with $D > 1$ exhibit a shear modulus, characterising the stiffness of the crystal to transverse (i.e. shear) deformations. This also manifests as a new gapless branch of transverse excitations, where the motion of the crystal is perpendicular to the direction of wave propagation. This is in contrast to the other two gapless branches which are longitudinal.

To illustrate the properties of $D = 2$ supersolids we present results for two systems. These systems differ in the relative importance of elastic and compressibility effects, and exhibit different behavior for first and second sound across the transition. The first system is a dipolar Bose-Einstein condensate (BEC) in a planar trap, where the atoms are confined along the direction that dipoles are polarized and free in the perpendicular plane. The second system is a 2D BEC of atoms interacting with a finite-ranged soft-core interaction. The excitations of both systems can be obtained by numerically solving the Bogoliubov-de Gennes (BdG) equations. While the excitation spectrum for the 2D soft-core system has been previously studied [27–29], we present the first results for 2D dipolar excitations in the thermodynamic regime. The thermodynamic limit has the advantage over finite (harmonically trapped) system studies (e.g. see [20, 30]), because the in-plane quasimomentum is a good quantum number, leading to well-defined excitation bands that allow a clear interpretation of their properties (e.g. speeds of sound). For both systems we develop a hydrodynamic model, involving five elastic parameters that we determine from ground state calculations. We show that the hydrodynamic model provides an accurate description of the speeds of sound determined by numerical calculation of the BdG excitations, and provides insight into the origin of the different behavior of the two system across the transition, and deep into the crystal regime. While both systems studied are not directly comparable (e.g., the dipolar BEC is a three-dimensional system, while the soft-core BEC is 2D) by studying both we reveal the general applicability of hydrodynamic theory to 2D supersolids.

The outline of the paper is as follows. In Sec. II and Appendices A and B we describe the two systems we study in this work, and their transition to a supersolid state with a triangular crystalline structure. We present results for the excitations and dynamic structure factors determined by numerical solutions

[1] These associated excitation branches are also commonly referred to as density and phase branches. The second sound is associated to a reduced superfluidity (i.e. normal component) arising from the spontaneously broken translational symmetry (see [12, 14, 15]).

of the BdG equations in Sec. III. In Sec. IV we outline the hydrodynamic theory for the $D = 2$ supersolid. We discuss how the relevant elastic parameters that appear in this theory can be determined from ground state calculations. The hydrodynamic predictions for the speeds of sound are compared to those obtained from the BdG calculations. We then discuss the key parameters distinguishing the behavior of the dipolar and soft-core systems and identify the two relevant limits of the hydrodynamic results to describe these systems. Finally, we conclude in Sec. V.

II. SUPERSOLID SYSTEMS

Here we introduce the two systems examined in this work. Both systems are described by Hamiltonians that are translationally invariant in the xy-plane. The ground state phase diagram depends on the average atomic areal density, ρ, and various microscopic parameters (e.g. interactions), with 2D crystalline ground states occurring in appropriate parameter regimes.

The first system we introduce is a dipolar BEC of highly magnetic atoms under axial harmonic confinement with angular frequency ω_z, but with the atoms free to move in the xy-plane. The atoms interact by a short ranged contact interaction with s-wave scattering length a_s and the long-ranged dipole-dipole interaction, characterized by the dipole length a_{dd}. In this system quantum fluctuation effects become important in the dipole dominant regime, $\epsilon_{dd} \equiv a_{dd}/a_s > 1$, with the fluctuations able to stabilise the condensate from mechanical collapse [31–33]. This system is well-described by an extended meanfield theory with details given in Appendix A. Here we consider a BEC of ^{164}Dy atoms with $\omega_z/2\pi = 72.4\,\text{Hz}$, and $a_{dd} = 130.8\,a_0$, to be comparable to the phase diagrams produced for this system in Refs. [18, 22]. The system is constrained to have mean areal density ρ, which serves as a thermodynamic parameter. We choose to present results for $\rho = 0.04/a_{dd}^2 \approx 8.3\times10^{14}\text{m}^{-2}$, which is well below the critical point density of $\rho_c \approx 0.098/a_{dd}^2$, and is comparable to the densities used in experiments [16]. Under these conditions the nature of the ground state depends on the value of a_s, which is controlled in experiments using Feshbach resonances. For our choice of density the ground state is uniform for $\epsilon_{dd} < 1.31$ (i.e. $a_s > 99.7a_0$) and is a triangular crystal for $\epsilon_{dd} \geq 1.31$. We show examples of these states in Fig. 1. As ϵ_{dd} increases the overlap between unit cells decreases and at each lattice site a filament shaped droplet (elongated in the dipole direction z) forms. It is useful to characterise the strength of the modulation of the density in the xy-plane by the density contrast defined as

$$C = \frac{\varrho_{\max} - \varrho_{\min}}{\varrho_{\max} + \varrho_{\min}}, \tag{1}$$

where $\varrho_{\max}$ and $\varrho_{\min}$ are the maximum and minimum of the areal density $\varrho(x, y)$ of the ground state. The density contrast as a function of ϵ_{dd} is shown in Fig. 1(a), revealing the discontinuous character of the transition. In these results we have extended the uniform superfluid state beyond the transition point where it is a metastable state. However, at $\epsilon_{dd} \approx 1.32$ ($a_s \approx 99.0a_0$), a roton excitation softens and the uniform state is dynamically unstable.

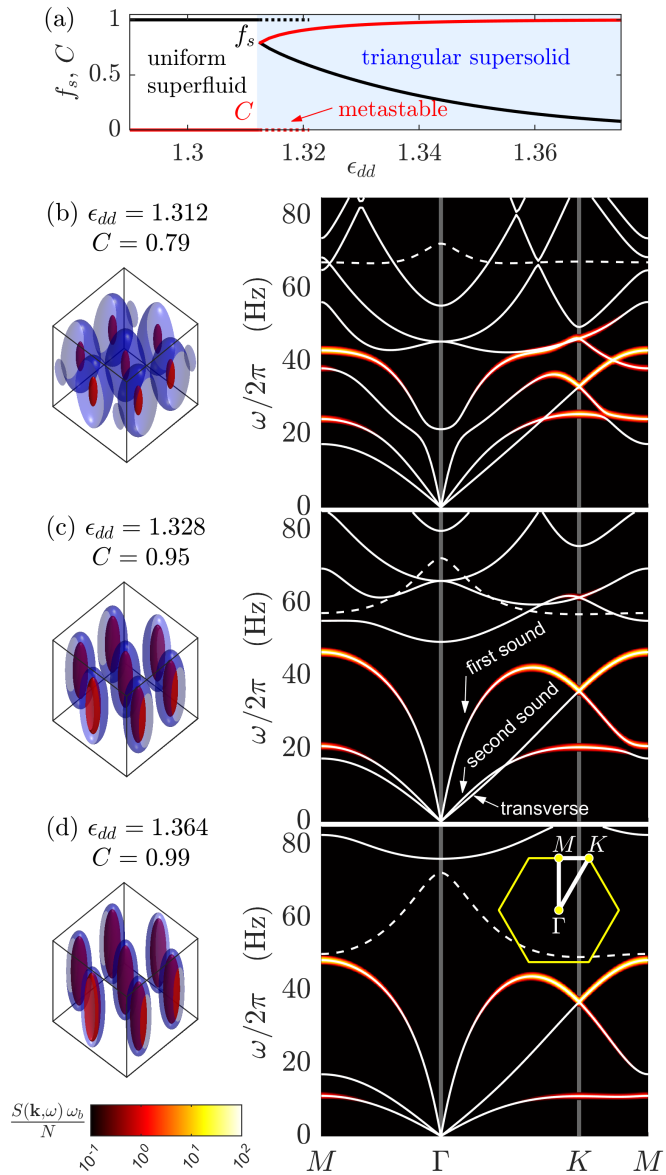

FIG. 1. (a) The superfluid fraction f_s and density contrast across the first-order uniform superfluid to supersolid phase transition. (b)-(d) Ground state density (left) and excitation spectra along the three symmetry directions of the Brillouin zone (right) [see inset to subplot (d)] for a planar dipolar gas at three values of ϵ_{dd}. Density isosurfaces at $0.55\times10^{20}\text{m}^{-3}$ (blue) and $3\times10^{20}\text{m}^{-3}$ (red) and shown in a box of size $11\,\mu\text{m}\times11\,\mu\text{m}\times14\,\mu\text{m}$. The heat map image shows the dynamical structure factor frequency broadened by $\omega_b = 10^{-2}\omega_z$. The excitation spectra are shown as white lines, dashed lines being odd axial modes which don't contribute to the structure factor. Results for ^{164}Dy atoms with an average areal density of $\rho = 0.04/a_{dd}^2$ and axial confinement of $\omega_z/2\pi = 72.4\,\text{Hz}$. The superfluid fraction is discussed in Sec. IV A.

The second system is a 2D BEC of atoms that interact via the soft-core potential of strength U_0 and range a_{sc}. This sys-

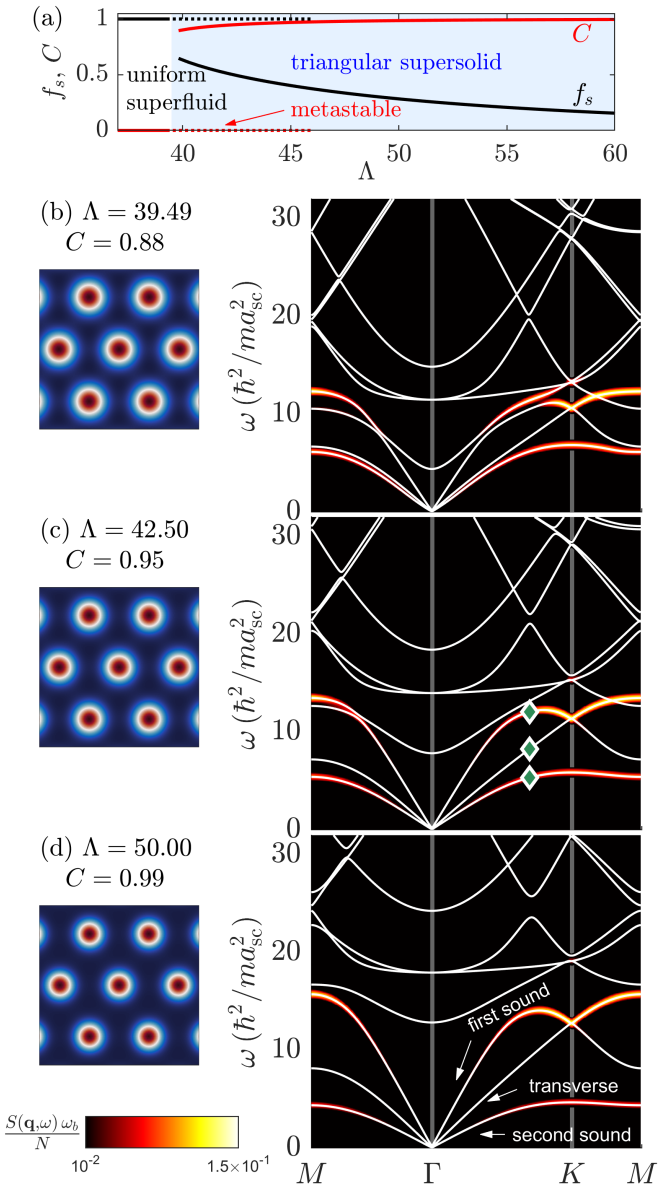

FIG. 2. (a) The superfluid fraction and density contrast across the first-order uniform superfluid to supersolid phase transition. (b)-(d) Ground state density (left) and excitation spectra along the three symmetry directions of the Brillouin zone (right) for a 2D soft-core system at three values of Λ. The ground state densities are shown in a box of size $4\,a_{sc} \times 4\,a_{sc}$. The excitation spectra are shown as white lines, on top of a heat map image showing the dynamical structure factor frequency broadened by $\omega_b = 0.35\omega_0$. The three symmetry directions of the Brillouin zone are the same as the ones depicted in the inset of Fig. 1(d). The three green diamonds mark the position of the modes plotted in Fig. 3 in an equivalent symmetry direction.

tem has been extensively studied as supersolid model (e.g. see [27–29, 34–38]). Schemes have been proposed to realize soft-core interactions in ultra-cold atom experiments [39, 40], but there has been limited reported experimental activity in the regime relevant to supersolidity thus far. We consider the system in a regime well-described by meanfield theory, with further details given in Appendix B. The phase diagram for the 2D soft-core model depends on the single dimensionless parameter $\Lambda = m\pi\rho a_{sc}^4 U_0/\hbar^2$. The melting value $\Lambda_m = 39.49$ separates uniform superfluid states (for $\Lambda < \Lambda_m$) from triangular crystal states. We show examples of these states in Fig. 2, and the behavior of the density contrast as a function of Λ is shown in Fig. 2(a). Similar to the dipolar system, the transition from unmodulated to modulates states also occurs discontinuously as Λ changes, with the unmodulated states remaining metastable until the roton completely softens at $\Lambda = 46.30$.

For both systems, we find the ground state by minimizing the energy density, $\mathcal{E}$, i.e. the energy per unit area. It is convenient to write this $\mathcal{E}(\rho; \mathbf{v}, \mathbf{a}_1, \mathbf{a}_2)$, being a function of ρ, the superfluid velocity[2] $\mathbf{v}$ and $\{\mathbf{a}_1, \mathbf{a}_2\}$, being the direct lattice vectors of the crystal. The ground state for average density ρ is obtained by minimising the energy density for $v = 0$ with respect to the lattice constant[3]. Additional details are given in Appendices A and B.

III. EXCITATION RESULTS

The excitation spectrum can be determined by linearizing around the ground state $\psi(\mathbf{x})$ with a time-dependent ansatz of the form

$$\Psi(\mathbf{x}, t) = e^{-i\mu t/\hbar}\left[\psi(\mathbf{x}) + \sum_{\nu\mathbf{q}}\left\{c_{\nu\mathbf{q}}u_{\nu\mathbf{q}}(\mathbf{x})e^{-i\omega_{\nu\mathbf{q}}t} - c^*_{\nu\mathbf{q}}v^*_{\nu\mathbf{q}}(\mathbf{x})e^{i\omega^*_{\nu\mathbf{q}}t}\right\}\right], \quad (2)$$

where μ is the ground state chemical potential, $c_{\nu\mathbf{q}}$ are the expansion amplitudes, and ν and $\mathbf{q}$ are the band index and planar quasimomentum of the excitation, respectively. Here $\{u_{\nu\mathbf{q}}(\mathbf{x}), v_{\nu\mathbf{q}}(\mathbf{x})\}$ are the quasiparticle amplitudes, with respective energies $\hbar\omega_{\nu\mathbf{q}}$, and these are determined by solving the BdG equations (see Appendix C, and Refs.[28, 41]).

In Figs. 1 and 2 we show results for the excitation spectra for the dipolar and soft-core models, respectively. These results are shown along the symmetry directions of the first Brillouin zone. For both sets of results the case shown in subplot (b) is close to the phase transition, whereas subplots (c) and (d) are for states with higher values of the density contrast. The excitations are shown as solid white lines on top of the dynamic structure factor, $S(\mathbf{q}, \omega)$, which is obtained as

$$S(\mathbf{q}, \omega) = \sum_{\nu}\left|\int d\mathbf{x}\,(u^*_{\nu\mathbf{q}} - v^*_{\nu\mathbf{q}})e^{i\mathbf{q}\cdot\mathbf{x}}\psi\right|^2 \delta(\omega - \omega_{\nu\mathbf{q}}). \quad (3)$$

The dynamic structure factor reveals the density fluctuations associated with the excitations, notably $S(\mathbf{q}, \omega)$ vanishes for excitations that do not affect the density.

[2] We are interested in stationary superfluids, but introduce $\mathbf{v}$ to define the superfluid fraction.

[3] The ground state configuration for both models is a triangular lattice, thus we can take $\mathbf{a}_1 = a\hat{\mathbf{x}}$, and $\mathbf{a}_2 = \frac{1}{2}a\hat{\mathbf{x}} + \frac{\sqrt{3}}{2}a\hat{\mathbf{y}}$, with lattice constant a.